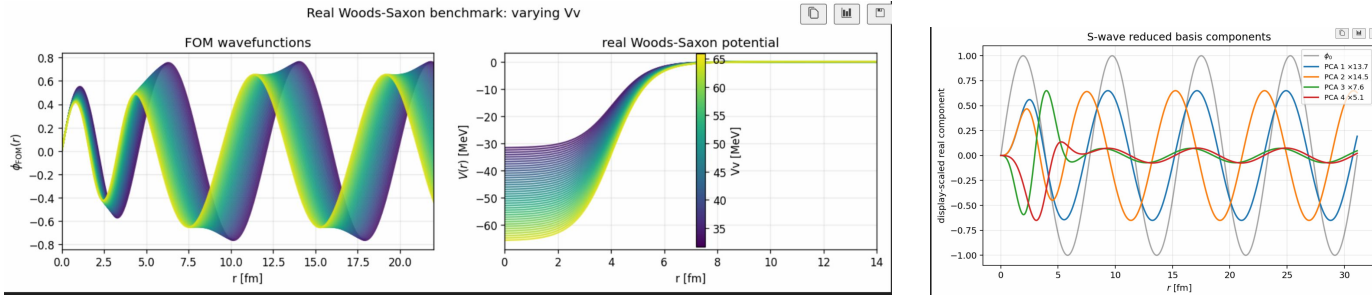


Notebook 1: Comparing traditional RBM to LROM for simplest case with one wavefunction only

1. Show current status of things in terms of RBMs for scattering, with simple example: $l=0$, WS real term with three parameters only.
 - a. Simple first case, varying V_v only:
 - i. Rainbow plot for potential and wavefunctions, show RB for wavefunction



We should work with the central solution for ϕ_0

Notebook 1: Comparing traditional RBM to LROM for simplest case with one wavefunction only

Show current status of things in terms of RBMs for scattering, with simple example: $l=0$, WS real term with three parameters only.

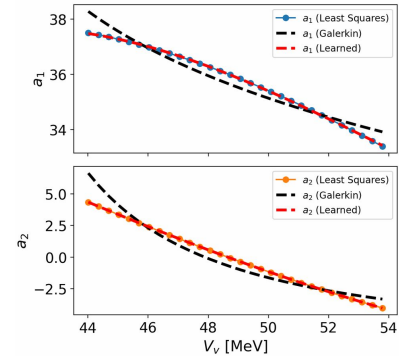
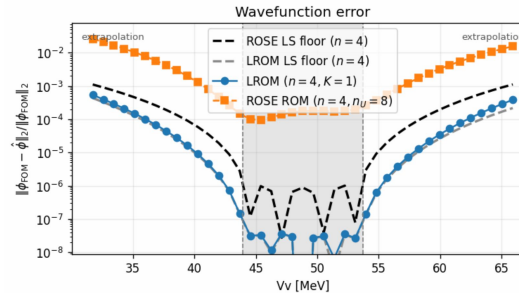
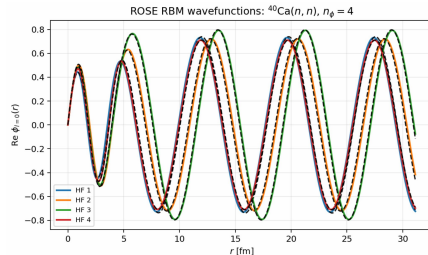
1. Simple first case, varying V_v only:

a. Rainbow plot for potential and wavefunctions, show RB for wavefunction

b. Show coefficients obtained with galerkin projection and with LROM:

$$\left(\mathbf{M}_0 + V_v \mathbf{M}_1\right) \vec{a} = -V_v \vec{c}$$

c. Show performance on reproducing the wavefunctions



Notebook 1: Comparing traditional RBM to LROM for simplest case with one wavefunction only

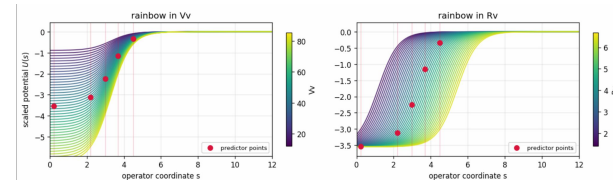
Show current status of things in terms of RBMs for scattering, with simple example: $l=0$, WS real term with three parameters only.

1. Simple first case, varying V_v only
2. More complicated case, varying three parameters V_v , R_v , and av . LROM equations need to be transformed to this:

$$\left[I + \sum_{j=1}^K p_j(\alpha) \mathbf{M}_j \right] \vec{a} = \sum_{j=1}^K p_j(\alpha) b_j$$

and RBM from ROSE should use EIM. LROM needs to have the ability to compute the maxvol algorithm for selecting predictors

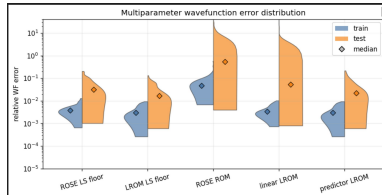
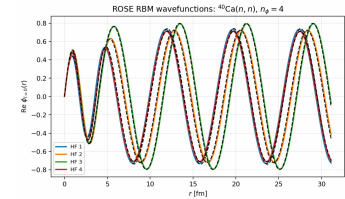
Add av rainbow plot too



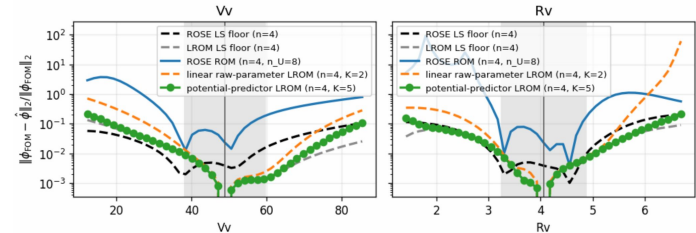
Notebook 1: Comparing traditional RBM to LROM for simplest case with one wavefunction only

Show current status of things in terms of RBMs for scattering, with simple example: $l=0$, WS real term with three parameters only.

1. Simple first case, varying V_v only
2. More complicated case, varying three parameters V_v , R_v , and a_v .
 - a. Show results for emulating wavefunction across parameter samples and compare to ROSE. Figures like these can help visualize performance in interpolation and extrapolation

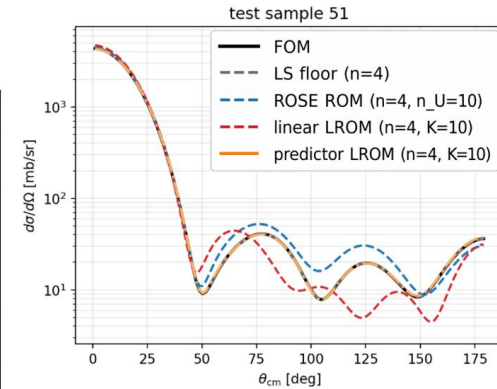
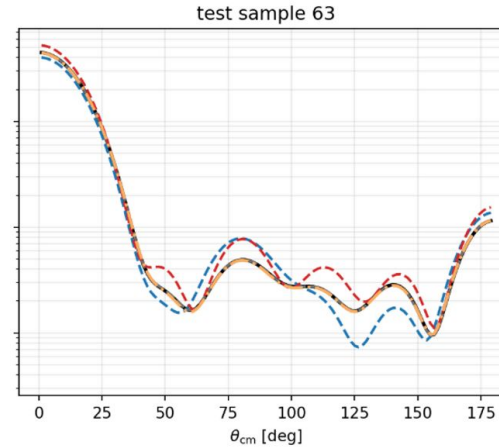
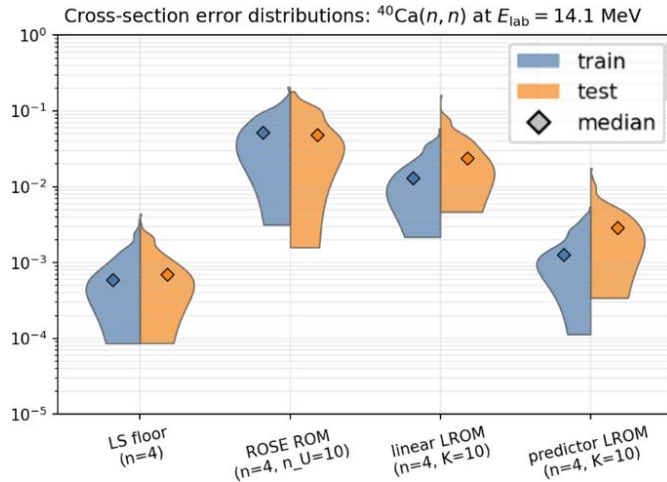


Dont show linear LROM, only LS floor, ROSE, and LROM



Notebook 2: Comparing ROSE and LROM at the cross section level when varying all parameters of optical potential from ROSE paper

1. Using the same training samples across the parameter space we build both ROSE and LROM and compare their abilities to predict cross sections for the same basis number:

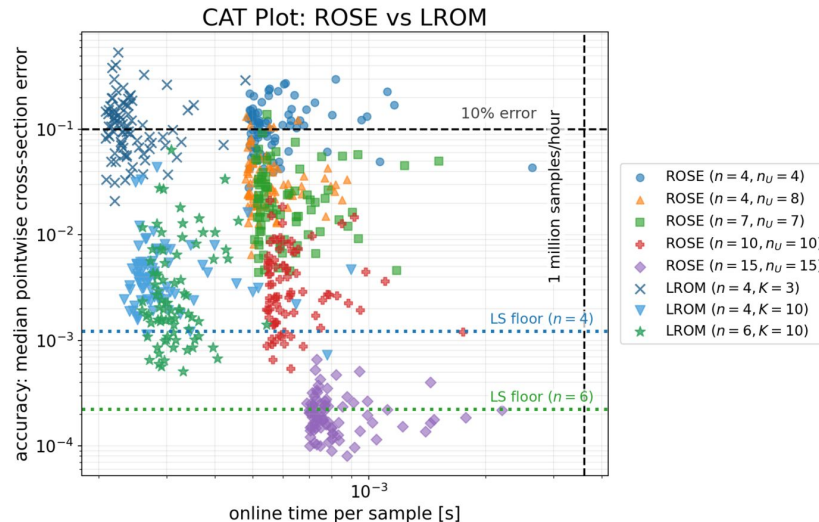


No need to show linear LROM

Can also show the rainbow plots and where the predictors land for some values of l , the nice plots you were showing in the call Daniel

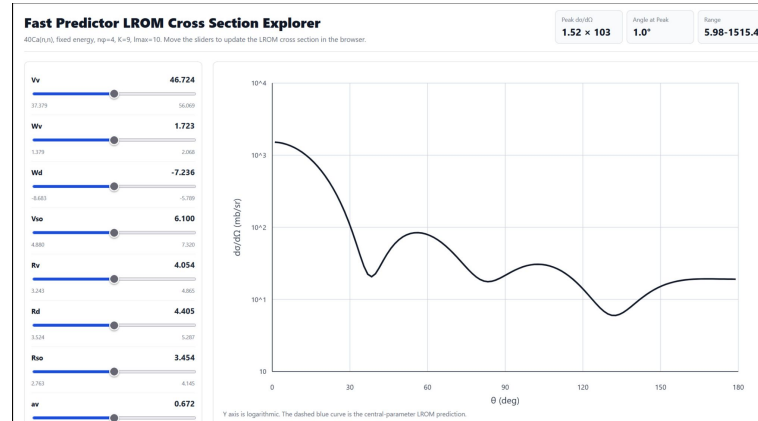
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1. Using the same training samples across the parameter space we build both ROSE and LROM and compare their abilities to predict cross sections for the same basis number
2. Compare in CAT plot space both ROSE and LROM for various basis and operator sizes:



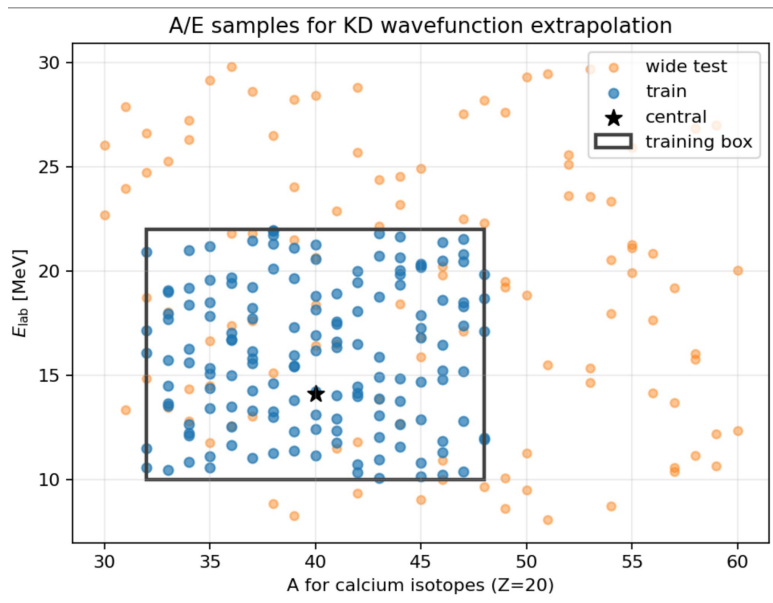
Notebook 2: Comparing ROSE and LROM at the cross section level when varying all parameters of optical potential from ROSE paper

1. Using the same training samples across the parameter space we build both ROSE and LROM and compare their abilities to predict cross sections for the same basis number
2. Compare in CAT plot space both ROSE and LROM for various basis and operator sizes
3. If possible (not sure how hard this is), have a part of the code that is able to transform a trained emulator into the html interactive file:



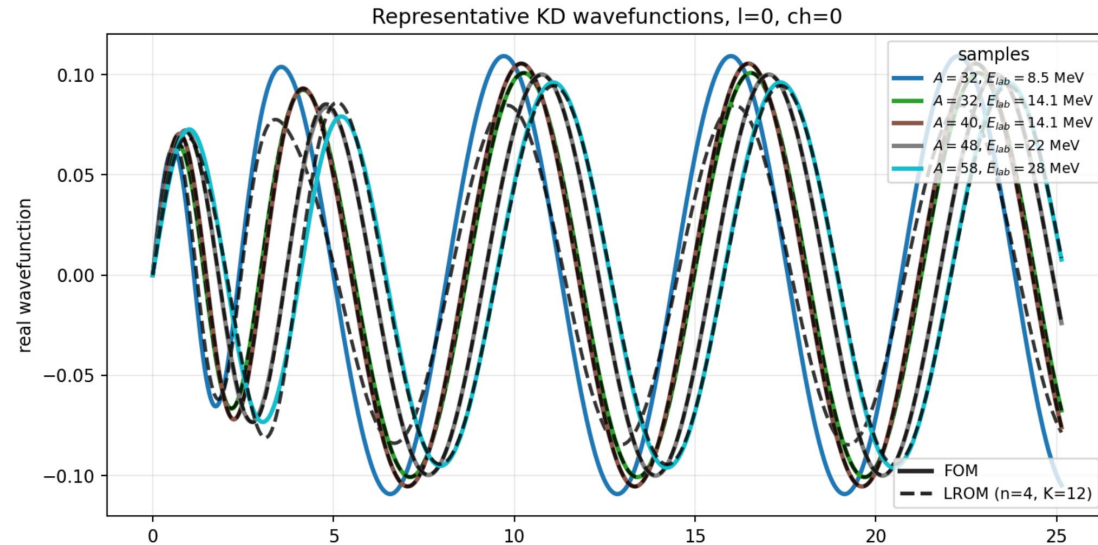
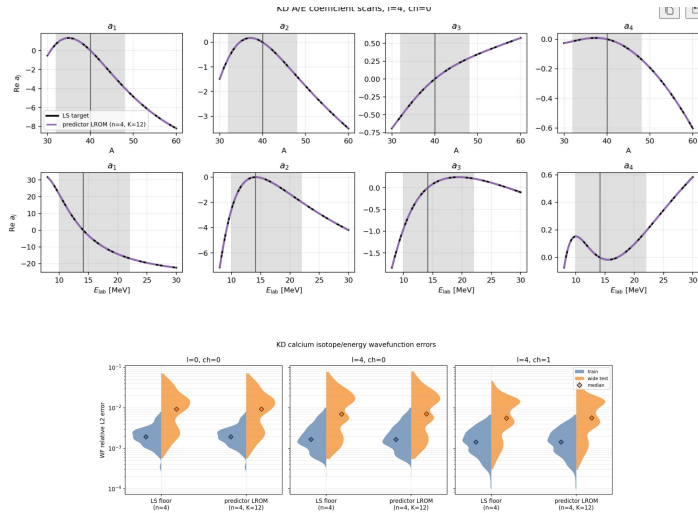
Notebook 3: Emulating a global optical potential that can calculate cross sections across isotopes and energies

1. Construct the FOM for the Koning-Delaroche parametrization, create training and testing samples more or less like this:



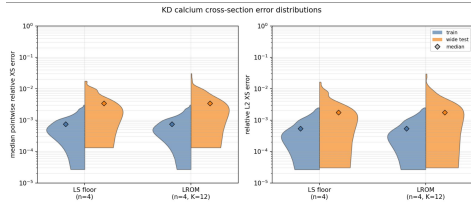
Notebook 3: Emulating a global optical potential that can calculate cross sections across isotopes and energies

1. Construct the FOM for the Koning-Delaroche parametrization, create training and testing samples more or less like this
2. Show results of emulating wavefunctions across all this, including coefficient performance and wavefunction performance

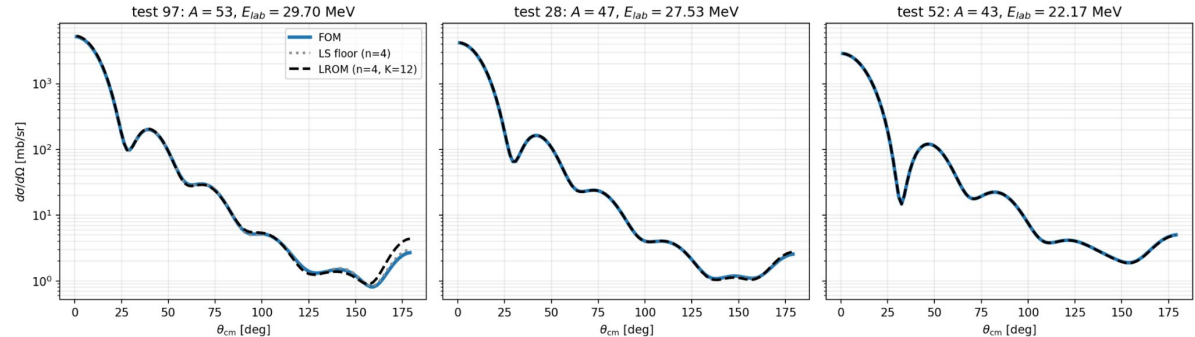


Notebook 3: Emulating a global optical potential that can calculate cross sections across isotopes and energies

1. Construct the FOM for the Koning-Delaroche parametrization, create training and testing samples more or less like this
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3. Show performance in cross sections in training and testing regions

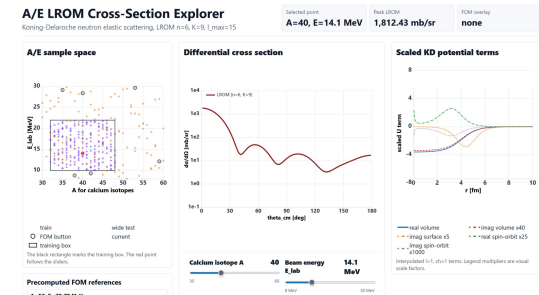


Representative KD calcium cross sections



Notebook 3: Emulating a global optical potential that can calculate cross sections across isotopes and energies

1. Construct the FOM for the Koning-Delaroche parametrization, create training and testing samples more or less like this
2. Show results of emulating wavefunctions across all this, including coefficient performance and wavefunction performance
3. Show performance in cross sections in training and testing regions
4. If possible (not sure how hard this is), have a part of the code that is able to transform a trained emulator into the html interactive file:



Notebook 3: Emulating a global optical potential that can calculate cross sections across isotopes and energies

1. Construct the FOM for the Koning-Delaroche parametrization, create training and testing samples more or less like this
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4. If possible (not sure how hard this is), have a part of the code that is able to transform a trained emulator into the html interactive file
5. Write something that is able to export the trained emulator as a callable function/black box

